SMART INDIA HACKATHON 2026

FDRFlow

SIH PS 26265 | Slide-by-Slide PPT Content Pack

Streamline of Security Deposit/FDR Portal

Secure. Trackable. Transparent.

Problem Statement ID	**SIH26265**
Organization	**AICTE**
Theme	**Smart Automation**
Category	**Software**

Team Name: [Enter your registered team name]
Team ID: [Enter your official SIH team ID]
Institute: [Enter your college / institute name]

Prepared as editable slide-ready content. Replace placeholders, mock figures and proposed targets with your actual team details and evidence.

SLIDE 1 | TITLE PAGE

Slide message Introduce FDRFlow as a secure, traceable portal for managing security-deposit/FDR records and their lifecycle.

On-slide text

FDRFlow

Streamline of Security Deposit/FDR Portal

Secure. Trackable. Transparent.

Field	Value
PS ID	SIH26265
Organization	AICTE
Theme	Smart Automation
Category	Software
Team Name	[Your registered team name]
Team ID	[Your official SIH team ID]
Institute	[Your college / institute]

Design direction

- Use a clean navy-blue, teal and white theme with a subtle visual of a digital document, bank deposit certificate and security shield.
- Place the SIH 2026 logo and team logo only if you have approved/official assets. Keep the title prominent and avoid long descriptions on this slide.

SLIDE 2 | PROPOSED SOLUTION

Solution name: FDRFlow

A centralized, role-based portal that manages each security deposit / Fixed Deposit Receipt (FDR) from submission through verification, maturity, renewal and authorized release.

A. Solution flowchart

1. Institution / Authorized User Login
Secure sign-in; role-based access.

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2. Submit FDR Details and Documents
Institution, bank/instrument reference, amount, issue date, maturity date and supporting files.

↓

3. Automated Completeness and Date Checks
Required-field validation, date consistency checks and duplicate-document flags.

↓

4. Authorized Verification and Official Scrutiny
Capture evidence and reviewer decisions; use supported external verification where available.

↓

5. Dashboard, Alerts and Workflow Tracking
Show status, pending actions, maturity dates, renewals and release requests.

↓

6. Updated Status, Audit Trail and Reports
Record status transitions, decisions, timestamps and responsible users.

B. Venn diagram: proposed data integration

Domain A: Institution Data	Domain B: FDR Data	Domain C: Workflow Data
Institute ID; approval status; institutional records	Bank/instrument reference; amount; issue and maturity dates	Verification; renewal; release requests; clearances; audit log
Institution identity and submission context	Financial instrument identity and lifecycle dates	Official actions and decision history

Centre of Venn diagram A ∩ B ∩ C = One Traceable FDR Record: a unified view of instrument details, verification state and administrative history.

Dataset note: These are proposed data domains, not a claim that AICTE publishes an open FDR dataset. Use synthetic or authorized anonymized records for the prototype.

C. USP / innovation

- Lifecycle tracking: connect submission, verification, maturity, renewal and release in one record.
- Evidence-linked verification: distinguish institution-submitted information from details independently confirmed by an authorized source.
- Proactive alerts: highlight approaching deadlines and pending actions.
- Transparent accountability: make status changes and scrutiny observations traceable.
- Human-controlled decisions: support officials without independently approving releases or declaring suspected fraud.

Product pitch: FDRFlow transforms fragmented FDR tracking into a secure, deadline-aware and audit-ready workflow.

SLIDE 3 | TECHNICAL APPROACH

A. Proposed technology stack

Layer / component	Technology	Purpose
Frontend	React.js	Responsive forms, dashboard, search, status timeline and role-specific views.
Backend runtime	Node.js	Runs server-side business logic.
API framework	Express.js	REST APIs, request validation and workflow endpoints.
Database	MySQL	Relational records for institutions, FDRs, workflows and audit events.
Charts	Chart.js	Dashboard visualizations and operational summaries.
Version control	Git + GitHub	Team collaboration, code review and release history.
Security / supporting services	JWT or secure session approach; role-based access; protected file storage; scheduled email alerts	Authentication, authorization, document handling and reminders.

B. System architecture

Users
Institution administrator; authorized AICTE officer; administrator/auditor.

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React frontend
Forms, dashboard, search, status timelines and reporting.

↓

Node.js + Express REST API
Authentication, permissions, validation and business rules.

↓

Data and document layer
MySQL for structured records; secure file storage for supporting documents.

↓

Supporting services
Alert scheduler and authorized external verification mechanism, when available.

↓

Audit and reporting
Status changes, review notes, timestamps and reports.

C. Live dashboard / prototype screenshot

Insert your actual dashboard screenshot here. Recommended visual: four KPI cards, a maturity timeline, pending-verification list, release/renewal status chart and recent activity/audit trail.

Illustrative KPI card	Demo value	Required caption
Active FDRs	128	Sample seed data
Expiring within 30 days	12	Sample seed data
Pending verification	9	Sample seed data
Release requests	7	Sample seed data

Data integrity The values above are mock figures for a possible UI, not official AICTE statistics or measured prototype results. Replace them with actual test output and label simulated records.

D. Links to demo assets

GitHub repository: [Insert repository URL]

Live prototype: [Insert deployed URL]

Demo video: [Insert video URL, if available]

Implementation safeguards

- Use server-side authorization and least-privilege access for each role.
- Validate uploaded files and store them in protected storage; avoid public links to financial documents.
- Connect bank verification only through authorized, supported mechanisms. Otherwise show “pending/manual review,” never a simulated success as a real verification.

SLIDE 4 | FEASIBILITY AND VISIBILITY

A. Feasibility assessment

Area	Why FDRFlow is feasible
Technical	Standard stack supports forms, databases, uploads, workflow and dashboards.
Operational	Workflow can reflect applicable AICTE steps and clearances.
Financial	Prototype can use open-source tools and synthetic data; production costs need assessment.
Integration	Use authorized APIs where available; otherwise mark manual review.
Security	Role-based access, protected storage and audit logs support safer handling.

B. Supporting facts and pilot targets

Reference	Fact and relevance
AICTE Approval Process Handbook	Lists FDR release documents and clearances; confirms the need for structured document and workflow tracking. Check applicable clauses.
Ministry of Education / PIB	Documents the 2024-2027 handbook and consultation process, providing context for configurable rules.

Links: AICTE Handbook | PIB announcement

Proposed pilot metrics (targets, not measured results)

Metric	Proposed target / definition
Record completeness	100% of active test records have required date/status fields.
Advance reminders	Configurable alerts 90, 30 and 7 days before maturity.
Workflow coverage	Five stages: submission, verification, maturity, renewal/replacement and release.
Manual effort	30% reduction is a target; measure against baseline before claiming.

Visibility and audience: Authorized AICTE/regional officers, institution administrators and auditors see relevant records, deadlines, pending actions and decision history. Use synthetic or approved anonymized pilot records.

C. Challenge–solution matrix

Challenge	FDRFlow response
Scattered FDR records	Central institution- and instrument-wise register.
Missing fields or mismatched dates	Required-field validation and discrepancy flags.
Missed maturity / renewal deadlines	Advance reminders and overdue alerts.
Unclear verification status	Pending/manual-review state plus evidence log.
Delayed release requests	Status workflow, clearance tracking and escalation.
Unauthorized access or record changes	Role-based access, protected files and audit logs.

SLIDE 5 | IMPACT AND BENEFITS

A. Expected benefits

Benefit area	Expected contribution
Social	Greater transparency for institutions, clearer responsibilities and fewer repetitive status enquiries.
Economic	Potential reduction in administrative effort, document retrieval costs and avoidable rework caused by missed follow-ups.
Technical	Unified records, controlled access, structured verification evidence and an audit-ready activity history.
Environmental	Less reliance on paper copies, physical filing and repetitive document correspondence.

B. Uniqueness compared with a basic manual workflow

The table compares FDRFlow with fragmented spreadsheet/email-based tracking, not with a verified audit of every feature in the existing official AICTE portal.

Capability	Manual spreadsheet / email	FDRFlow proposal
Centralized FDR register	No (typically fragmented)	Yes
Automatic maturity reminders	No (unless separately configured)	Yes
Instrument-wise status timeline	Partial / manual	Yes
Renewal and release workflow	Depends on existing process	Yes
Role-based dashboard	Depends on existing tools	Yes
Traceable scrutiny / action history	Partial / manual	Yes
Authorized bank verification, where supported	Depends on arrangements	Integrate where authorized
Final decision by competent authority	Yes	Yes

Legend: “Yes” means a proposed capability. “Partial / manual” means it depends on the local workflow. Validate these comparisons with actual users before presenting them as measured baseline findings.

C. Product-to-impact story

Stage	Narrative
Before: fragmented tracking	A staff member searches spreadsheets and email threads for FDR documents, follows up for missing clearances and checks maturity dates separately.
With FDRFlow	The institution submits one record. Validation identifies missing information; authorized officers capture verification and clearances; configured alerts highlight deadlines.
Impact to measure	Track manual handling time, record completeness, missed reminders, pending-case duration and document-retrieval time before and after a pilot.

Closing message: FDRFlow makes security-deposit management more transparent, traceable and proactive while preserving authorized financial verification and regulatory decision-making.

SLIDE 6 | RESEARCH AND REFERENCES

A. Research references

Research topic	Reference	Fact / contribution
SIH problem statement	Official SIH 2026 problem-statement portal	Verify the PS ID, official title, category and submission requirements.
Detailed PS 26265 description	SIH26265 - FDR Portal listing	Secondary listing describing centralized repository, maturity tracking, verification, release workflow and audit trail; cross-check against official portal.
AICTE regulatory process	AICTE Approval Process Handbook	Reference for applicable FDR release documents and clearances. Confirm the latest applicable clauses.
Regulatory framework	PIB: AICTE Approval Process Handbook 2024-2027	Government announcement on handbook scope and consultation process.
Security / authorization	OWASP Authorization Cheat Sheet	Supports server-side authorization and least-privilege design.
Document upload security	OWASP File Upload Cheat Sheet	Guidance on file validation and protected document storage.

Research integrity note: No public sample dataset should be assumed to exist for actual AICTE security deposits. Use synthetic or explicitly authorized anonymized records. A mock bank connection must not be presented as real verification.

B. Contribution table - six team members

Member	Primary responsibility	Contribution / deliverable
Member 1 - [Name]	Team lead and research	Problem analysis, requirements, regulatory research and coordination.
Member 2 - [Name]	UI/UX and frontend	Dashboard, login screens, FDR forms and responsive interface.
Member 3 - [Name]	Backend and APIs	REST APIs, business logic, workflow states and validation.
Member 4 - [Name]	Database and document management	MySQL schema, relationships, file storage and retrieval.
Member 5 - [Name]	Security and automation	Role-based access, audit trail, alerts and verification workflow.
Member 6 - [Name]	Testing and presentation	Test cases, dashboard validation, demo documentation and PPT/video.

Final statement: FDRFlow supports transparent, traceable and proactive FDR management without replacing authorized financial verification or regulatory decisions.

FINAL SUBMISSION CHECKLIST

- Confirm the official PS title, theme and category; enter the registered team name, team ID, institute and member names.
- Replace repository, live prototype, screenshot and demo-video placeholders with real assets and working links.
- Clearly label mock dashboard values as sample data and proposed performance figures as targets unless they have been measured.
- Check current AICTE handbook clauses; do not imply bank verification or regulatory approval unless it is genuinely authorized and completed.